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St. Joseph's College of Arts & Science (Autonomous)
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MT511- COMPLEX ANALYSIS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Find $\text{Arg}\left(1+\sqrt{3}i\right)$.

2. Write the function $f(z)=z^3+z+1$ in the form $f(z)=u(x,y)+iv(x,y)$.

3. Verify $f(z)=3x+y+i(3y-x)$ is entire.

4. Define Branch cut.

5. Evaluate $\int_1^2\left(\frac{1}{t}-i\right)^2 dt$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Reduce $\frac{5i}{(1-i)(2-i)(3-i)}$ to a real number.

7. If $f(z)=\frac{iz}{2}$ in the disk $|z|<1$ then show that $\lim_{z\to 1} f(z)=\frac{i}{2}$.

8. Show that $f(z)=|z|^2$ cannot have a derivative at any non zero point.

9. Show that $\cosh^2 z - \sinh^2 z = 1$.

10. If $a \le b$ and $w(t)$ is any complex valued function then show that $\left|\int_a^b w(t)dt\right| \le \int_a^b \left|w(t)\right|dt$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) i) Sketch the set of points determined by the condition $|z-1+i|=1$ where z be any complex number.

ii) Show that $-1-i = \sqrt{2}\exp\left[i\left(-\frac{3\pi}{4}+2n\pi\right)\right], n = 0, \pm1, \pm2, ...$

(or)

b) Find n[th] root of unity.

12. a) Discuss the transformation $w = z^2$.

(or)

b) Suppose that $\int(z) = u(x,y) + i\upsilon(x,y), z_0 = x_0 + iy_0$ and $w_0 = u_0 + i\upsilon_0$. Then prove that $\lim_{z\to z_0} f(z) = w_0$ if and only if $\lim_{(x,y\to(x_0,y_0))} u(x,y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} \upsilon(x,y) = \upsilon_0$.

13. a) Suppose that $f(z) = u(x,y) + i\upsilon(x,y)$ and that $f'(z)$ exist at the point $z_0 = x_0 + iy_0$. Then prove that the first order partial derivative of u and υ must exist at (x_0, y_0), and they satisfy the Cauchy-Riemann equation $u_x = \upsilon_y, u_y = -\upsilon_x$ there. Also prove that $f'(z_0)$ can be written as $f'(z_0) = u_x + i\upsilon_x$ where these partial derivatives are to be evaluated at (x_0, y_0).

(or)

b) Suppose that the function f is analytic in some domain D which contains a segment of the x axis and whose lower half is the reflection of the upper half with respect to that axis. The prove that $\overline{f(z)} = f(\overline{z})$ for each point z in the domain if and only if $f(x)$ is real for each point x on the segment.

14. a) i) Show that $\sin z = 0$ if and only if $z = n\pi, n = 0, \pm 1, \pm 2, \ldots$

ii) Show that $\frac{d}{dz}(\sin z) = \cos z$ for any complex no z.

(or)

b) Show that $2\sin z_1 \cos z_2 = \sin(z_1 + z_2) + \sin(z_1 - z_2)$.

15. a) Let C denotes the contour $z = z(t), a \le t \le b,$ and for any nonnegative constant M such that the values of f on C Satisfy the condition $|f(z)| \le M$. Then prove that $\left|\int_C f(z)\,dz\right| \le ML$ where L is the arc length of C.

(or)

b) i) Using Cauchy-Goursat theorem, show that $\int_C f(z)dz = 0$ when the contour C is the circle $|z| = 1$, in the either direction and, when $f(z) = \frac{z^2}{z-3}$.

ii) Let C be the positively oriented circle $|z| = 2$, using Cauchy integral formula, evaluate $\int_C \frac{z\,dz}{(9 - z^2)(z + i)}$.

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